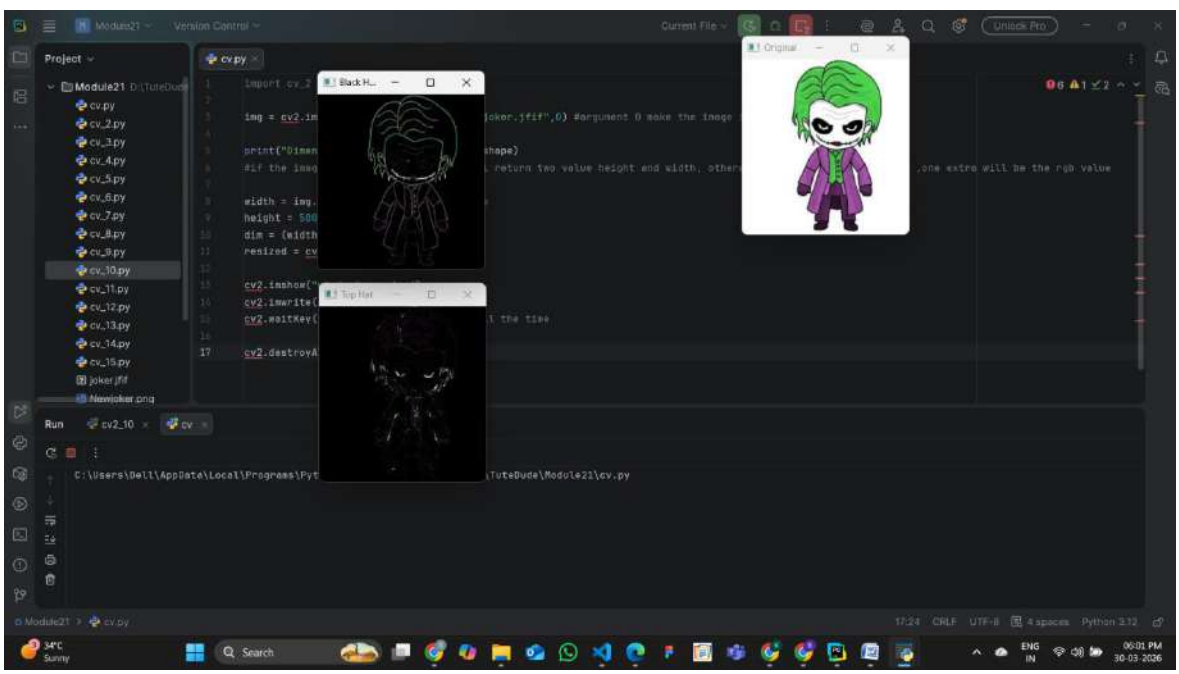
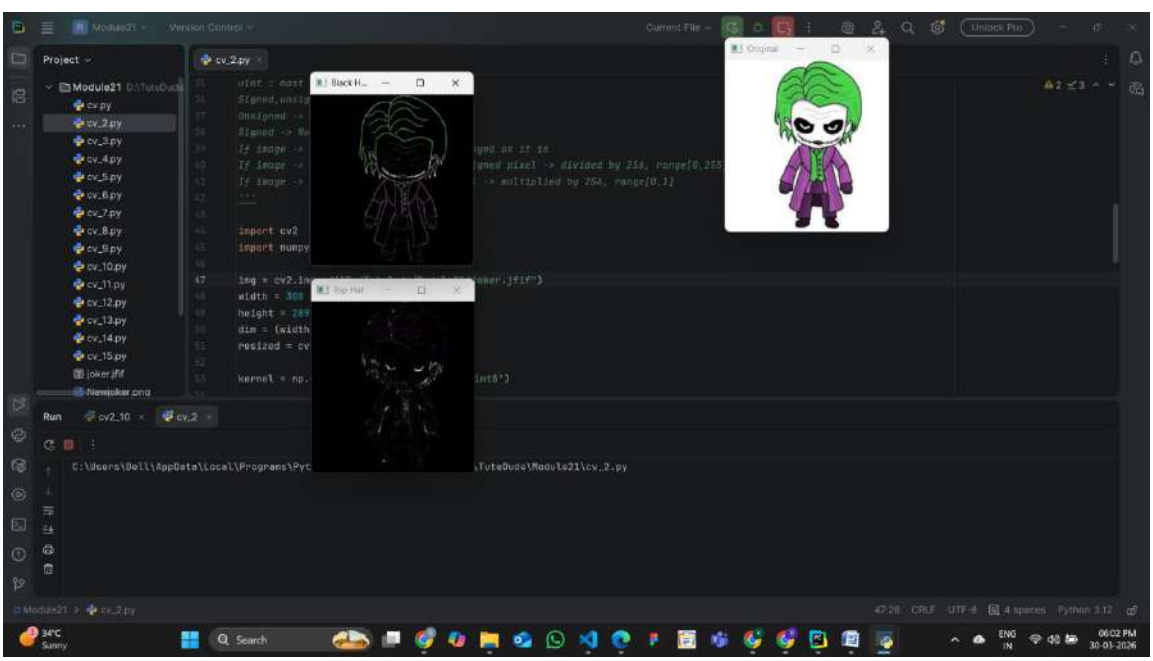
1. cv_1.py

```
import cv_2

img = cv2.imread("D:/TuteDude/Module21/joker.jfif",0) #argument 0 make
the image in black and white

print("Dimensions of the image : ",img.shape)
#if the image is black and white it will return two value height and
width, otherwise it will return 2 values ,one extra will be the rgb
value

width = img.shape[1] #width remains same
height = 500
dim = (width, height)
resized = cv2.resize(img, dim)

cv2.imshow("window", resized)
cv2.imwrite("Newjoker.png",img)
cv2.waitKey(1000) #window stays awake all the time

cv2.destroyAllWindows()
```



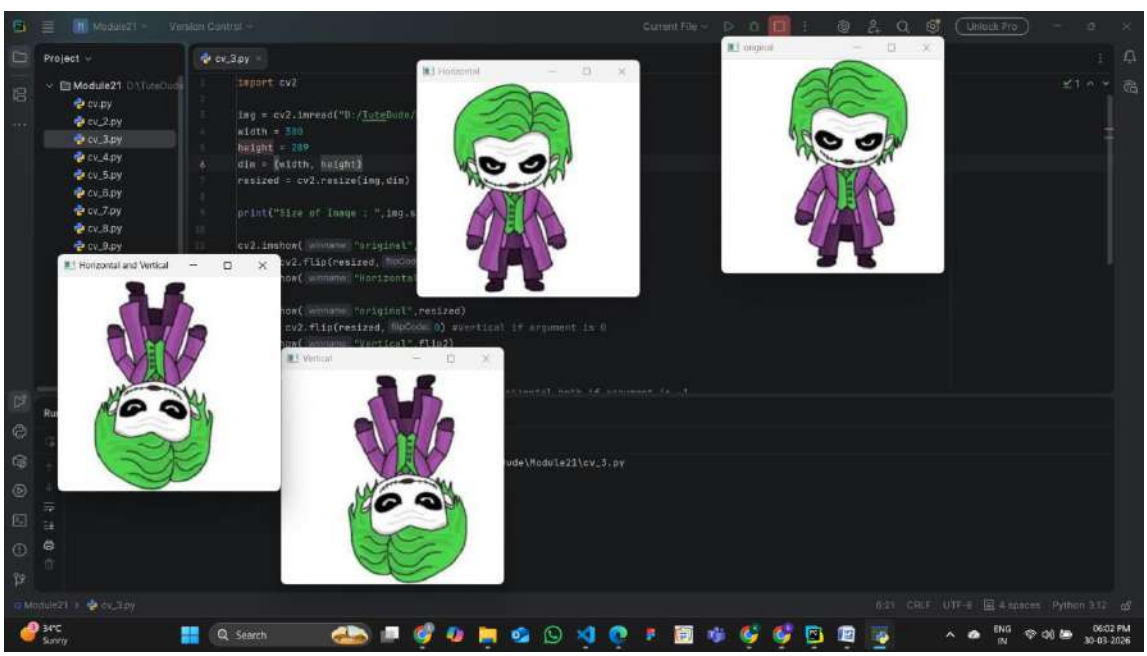
2. cv_2.py

```
import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")
width = 300
height = 289
dim = (width, height)
resized = cv2.resize(img,dim)

kernel = np.ones((3,3),dtype = 'uint8')

# erosion = cv2.erode(resized,kernel,iterations = 1)
# dilation = cv2.dilate(resized,kernel,iterations = 1)
# opening = cv2.morphologyEx(img,cv2.MORPH_OPEN,kernel)
# closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE,kernel)
# gradient = cv2.morphologyEx(img,cv2.MORPH_GRADIENT,kernel)
top_hat = cv2.morphologyEx(img,cv2.MORPH_TOPHAT,kernel)
black_hat =cv2.morphologyEx(img,cv2.MORPH_BLACKHAT,kernel)

cv2.imshow("Original",img)
# cv2.imshow("Erosion",erosion)
# cv2.imshow("Dilation",dilation)
# cv2.imshow("Opening",opening)
# cv2.imshow("Closing",closing)
# cv2.imshow("Gradient",gradient)
cv2.imshow("Top Hat",top_hat)
cv2.imshow("Black Hat",black_hat)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



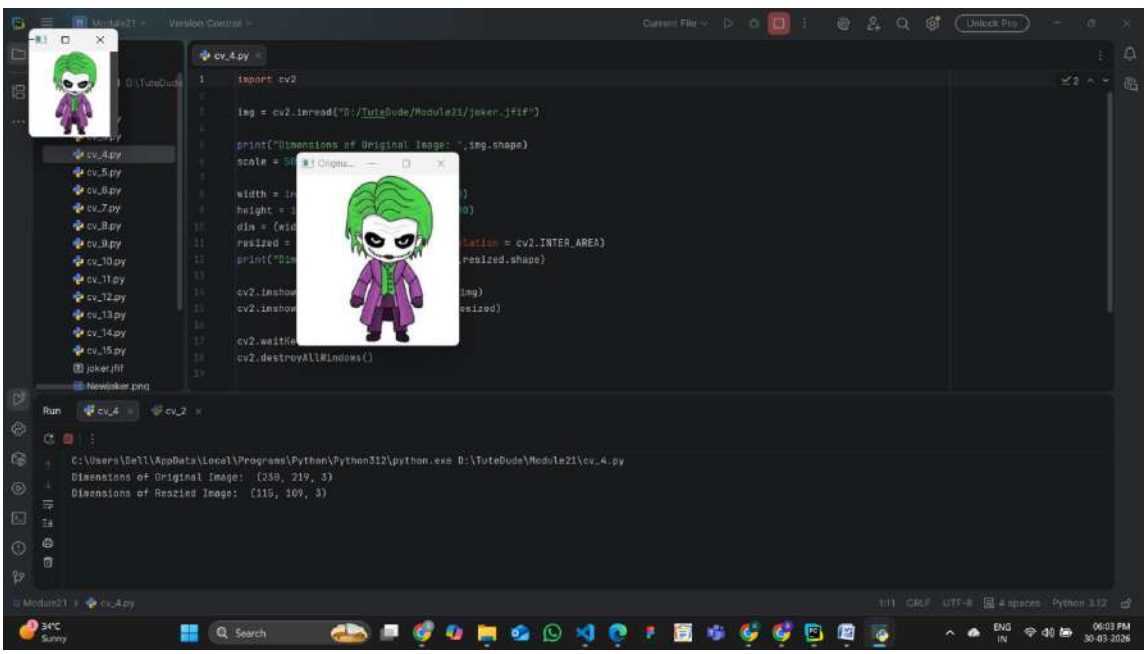
3. cv_3.py

```
import cv2

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")
width = 300
height = 289
dim = (width, height)
resized = cv2.resize(img,dim)

print("Size of Image : ",img.size) # in bytes

cv2.imshow("original",resized)
flip = cv2.flip(resized,1) #Horizontal if argument is 1
cv2.imshow("Horizontal",flip)

cv2.imshow("original",resized)
flip2 = cv2.flip(resized,0) #Vertical if argument is 0
cv2.imshow("Vertical",flip2)

cv2.imshow("original",resized)
flip3 = cv2.flip(resized,-1) #vertical and horizontal both if argument is
-1
cv2.imshow("Horizontal and Vertical",flip3)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



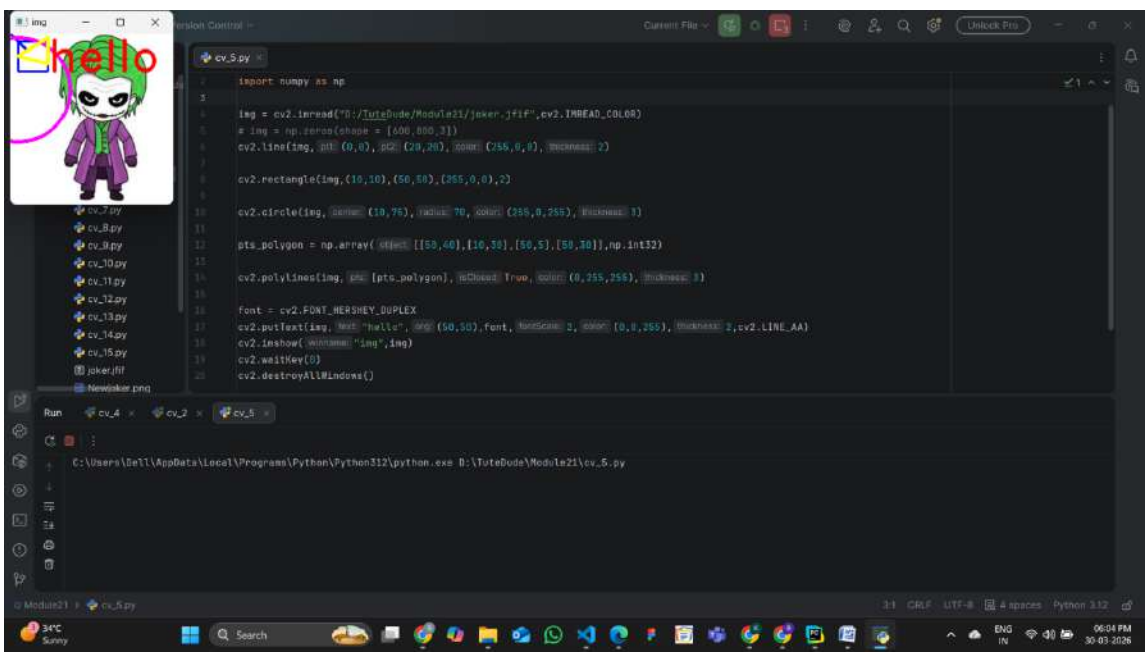
4. cv_4.py

```
import cv2

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")

print("Dimensions of Original Image: ",img.shape)
scale = 50

width = int(img.shape[1] * scale / 100)
height = int(img.shape[0] * scale / 100)
dim = (width, height)
resized = cv2.resize(img, dim, interpolation = cv2.INTER_AREA)
print("Dimensions of Reszied Image: ",resized.shape)

cv2.imshow("Original Image", img)
cv2.imshow("Resized Image", resized)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



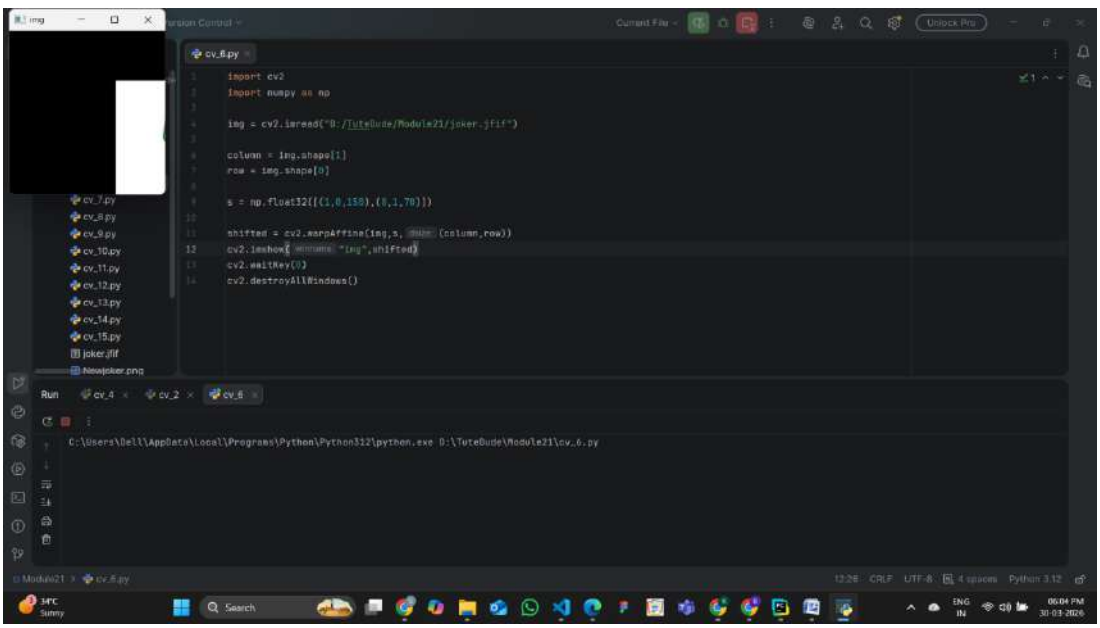
5. cv_5.py

```
import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif",cv2.IMREAD_COLOR)
# img = np.zeros(shape = [600,800,3])
cv2.line(img,(0,0),(20,20),(255,0,0),2)

cv2.rectangle(img,(10,10),(50,50),(255,0,0),2)

cv2.circle(img,(10,75),70,(255,0,255),3)

pts_polygon = np.array([[50,40],[10,30],[50,5],[50,30]],np.int32)

cv2.polylines(img,[pts_polygon],True,(0,255,255),3)

font = cv2.FONT_HERSHEY_DUPLEX
cv2.putText(img,"hello",(50,50),font,2,(0,0,255),2,cv2.LINE_AA)
cv2.imshow("img",img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



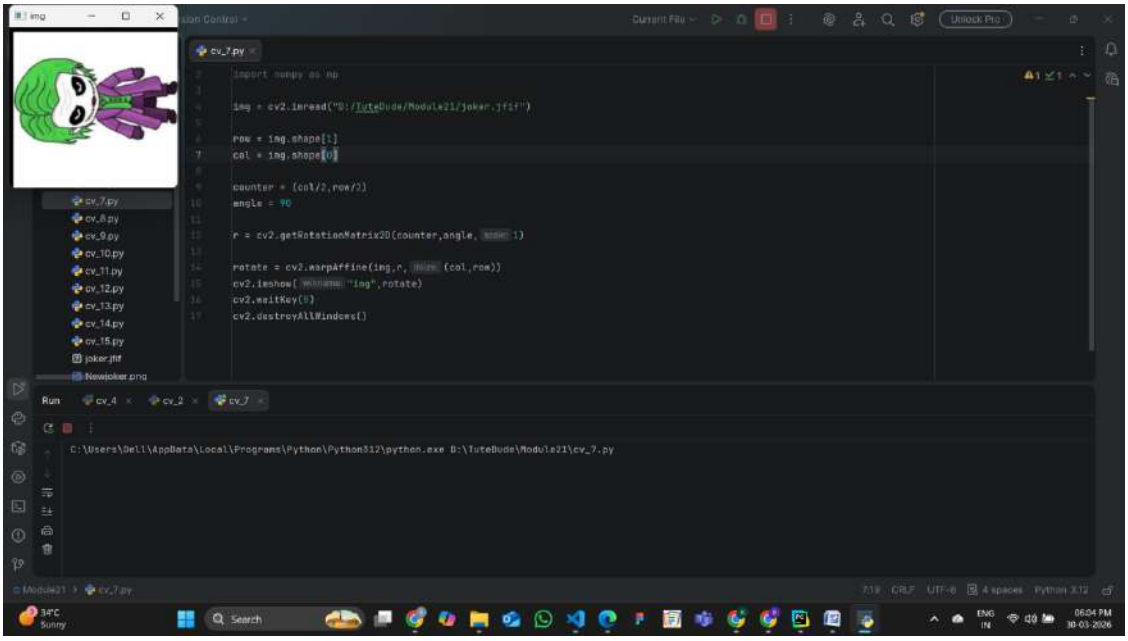
6. cv_6.py

```
import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")

column = img.shape[1]
row = img.shape[0]

s = np.float32([(1,0,150),(0,1,70)])

shifted = cv2.warpAffine(img,s,(column,row))
cv2.imshow("img",shifted)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



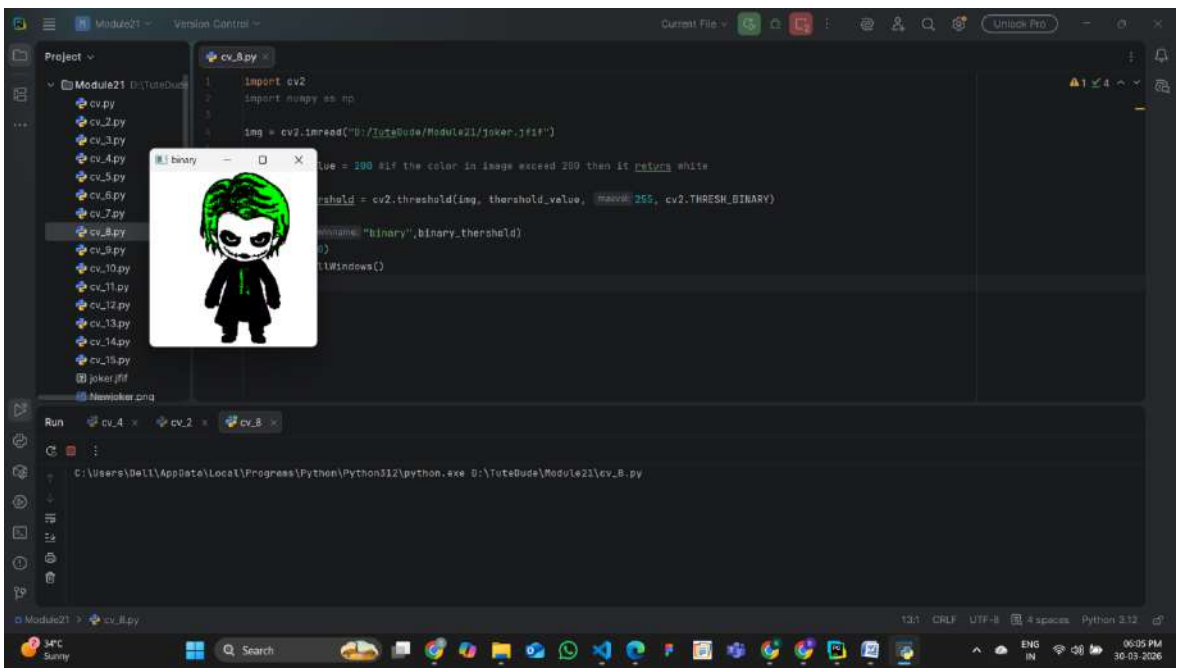
7. cv_7.py

```
import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")

row = img.shape[1]
col = img.shape[0]

counter = (col/2,row/2)
angle = 90

r = cv2.getRotationMatrix2D(counter,angle,1)

rotate = cv2.warpAffine(img,r,(col,row))
cv2.imshow("img",rotate)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



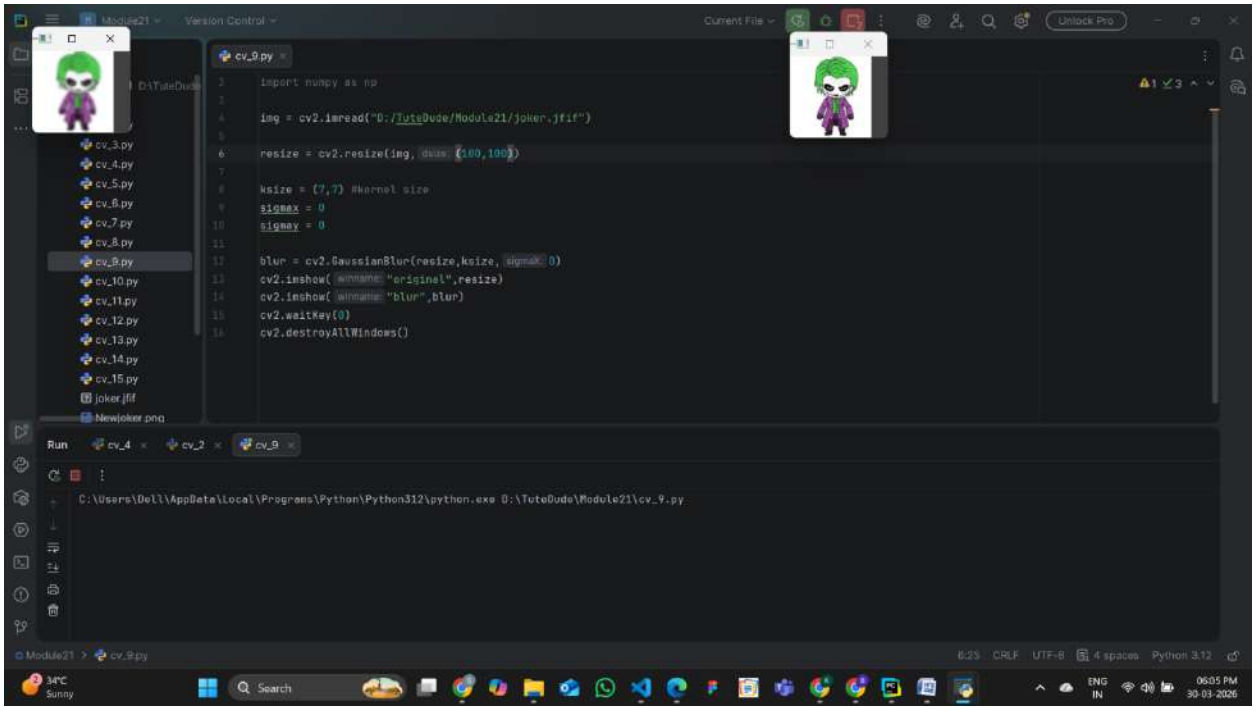
8. cv_8.py

```
import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")

thershold_value = 200 #if the color in image exceed 200 then it returs
white

_,binary_thershold = cv2.threshold(img, thershold_value, 255,
cv2.THRESH_BINARY)

cv2.imshow("binary",binary_thershold)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



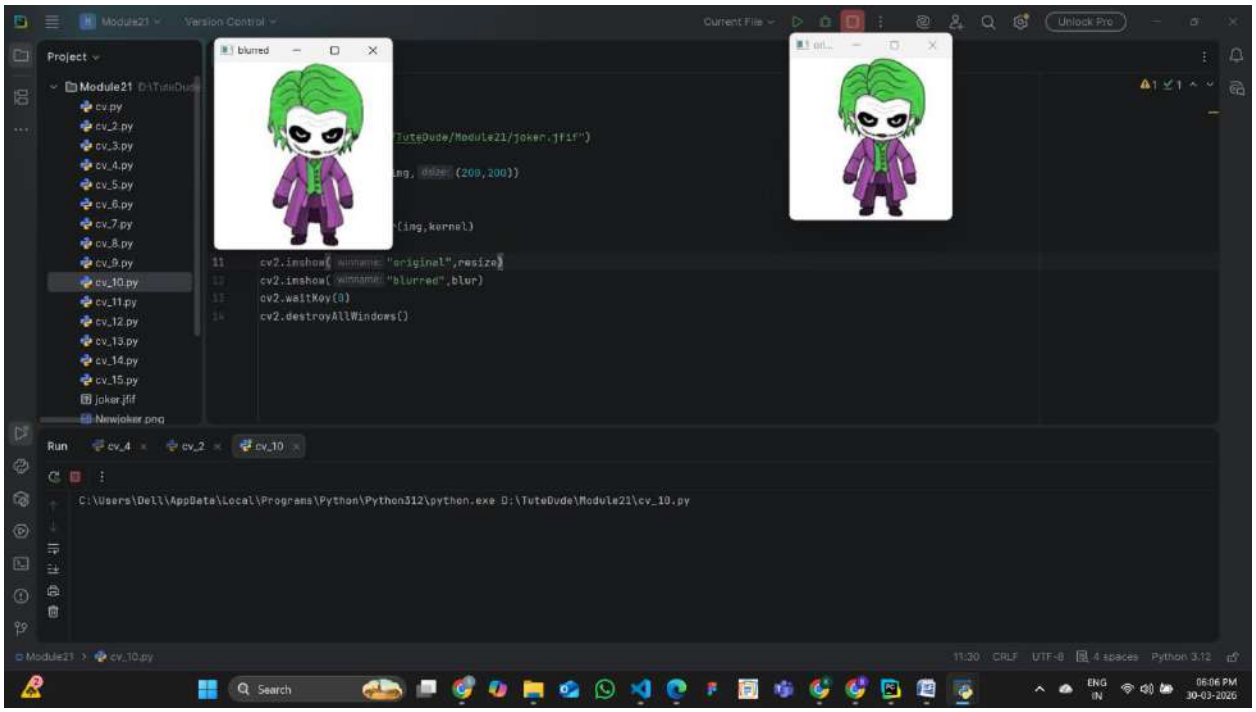
9. cv_9.py

```
import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")

resize = cv2.resize(img,(100,100))

ksize = (7,7) #kernel size
sigmax = 0
sigmay = 0

blur = cv2.GaussianBlur(resize,ksize,0)
cv2.imshow("original",resize)
cv2.imshow("blur",blur)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



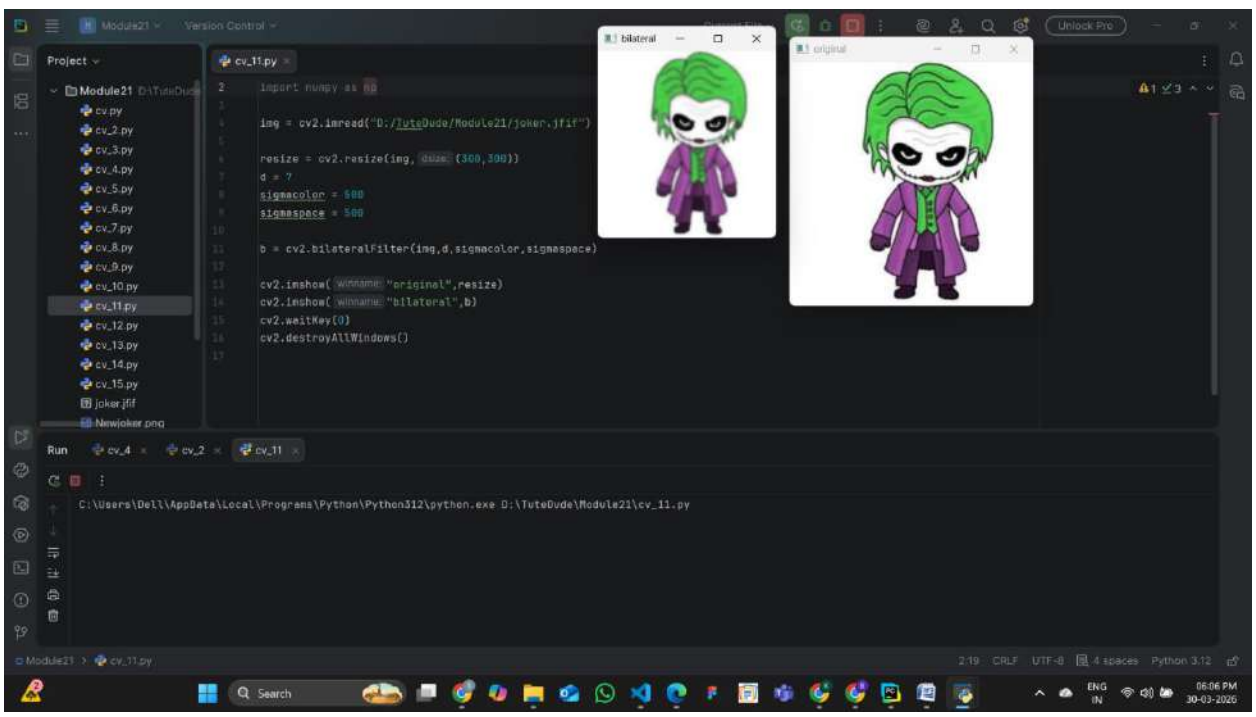
10. cv_10.py

```
import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")

resize = cv2.resize(img,(200,200))
kernel = 3

blur = cv2.medianBlur(img,kernel)

cv2.imshow("original",resize)
cv2.imshow("blurred",blur)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



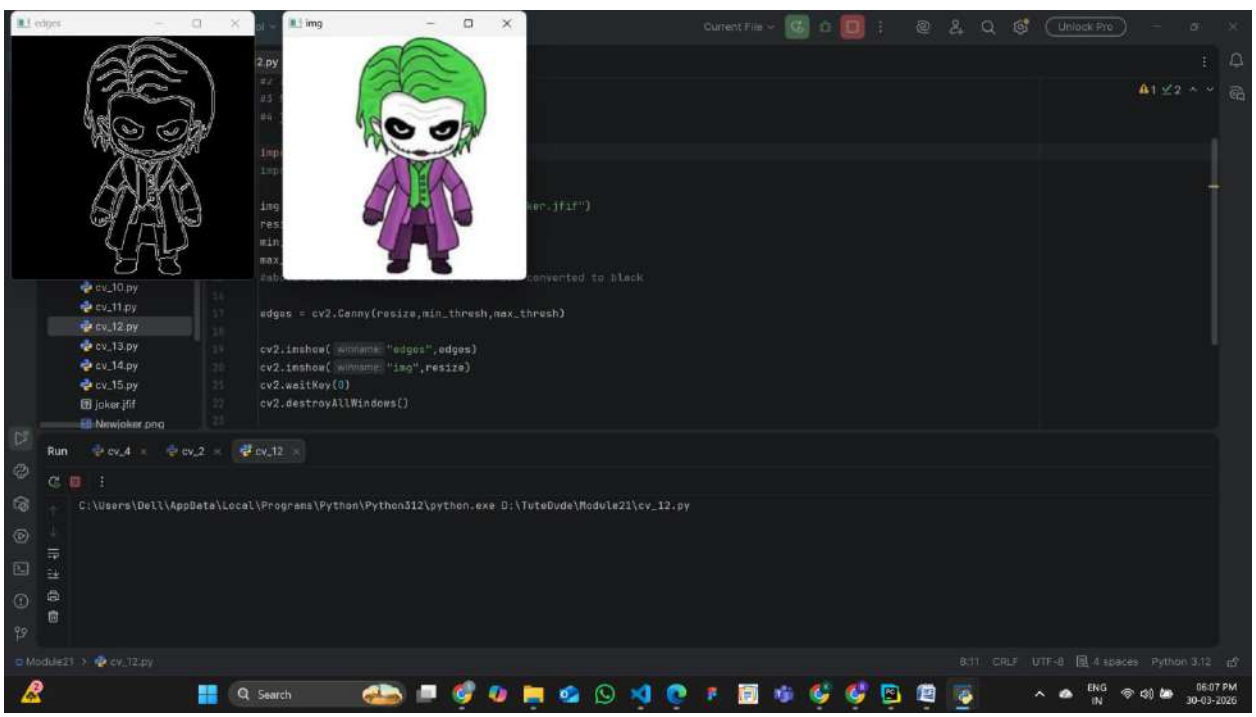
11. cv_11.py

```
import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")

resize = cv2.resize(img,(300,300))
d = 7
sigmacolor = 500
sigmaspace = 500

b = cv2.bilateralFilter(img,d,sigmacolor,sigmaspace)

cv2.imshow("original",resize)
cv2.imshow("bilateral",b)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



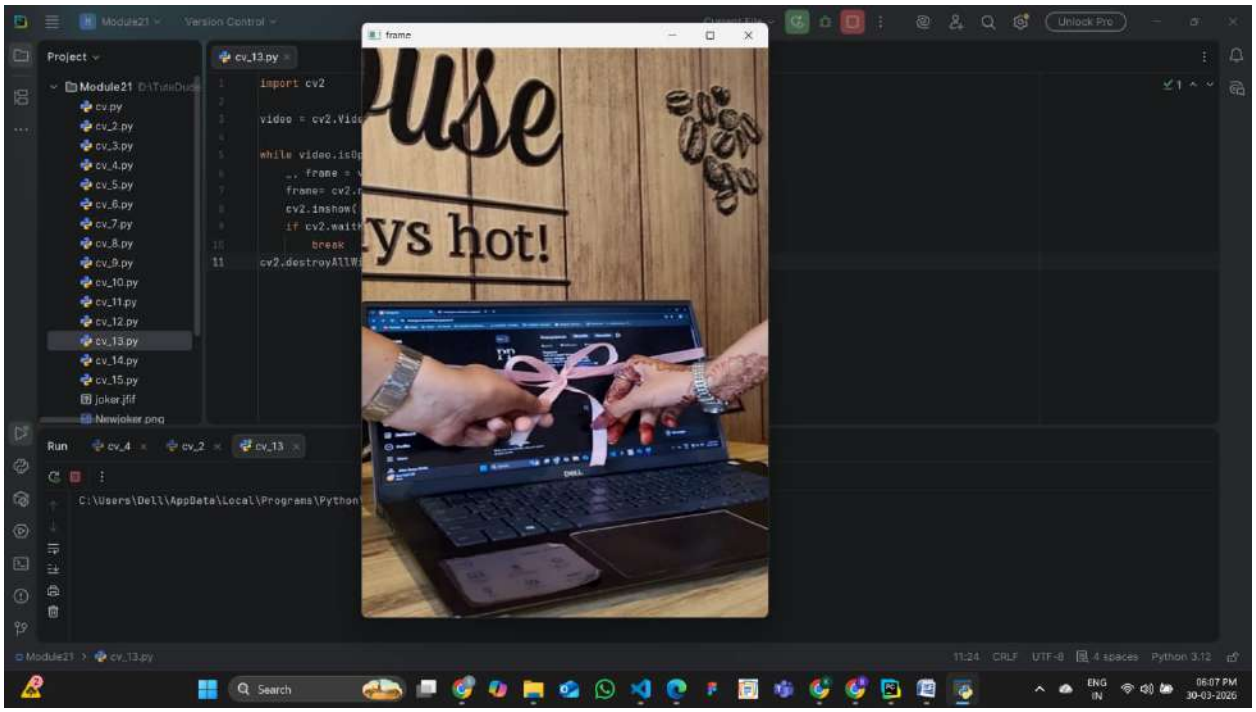
12. cv_12.py

```
#canny Edge Detection

#1 Noise reduction
#2 Intensity of the gradient of the image
#3 Non-maximum suppression
#4 Thersholding

import cv2
import numpy as np

img = cv2.imread("D:/TuteDude/Module21/joker.jfif")
resize = cv2.resize(img,(300,300))
min_thresh = 100
max_thresh = 200
#above 200 converted to white, below 100 converted to black

edges = cv2.Canny(resize,min_thresh,max_thresh)

cv2.imshow("edges",edges)
cv2.imshow("img",resize)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



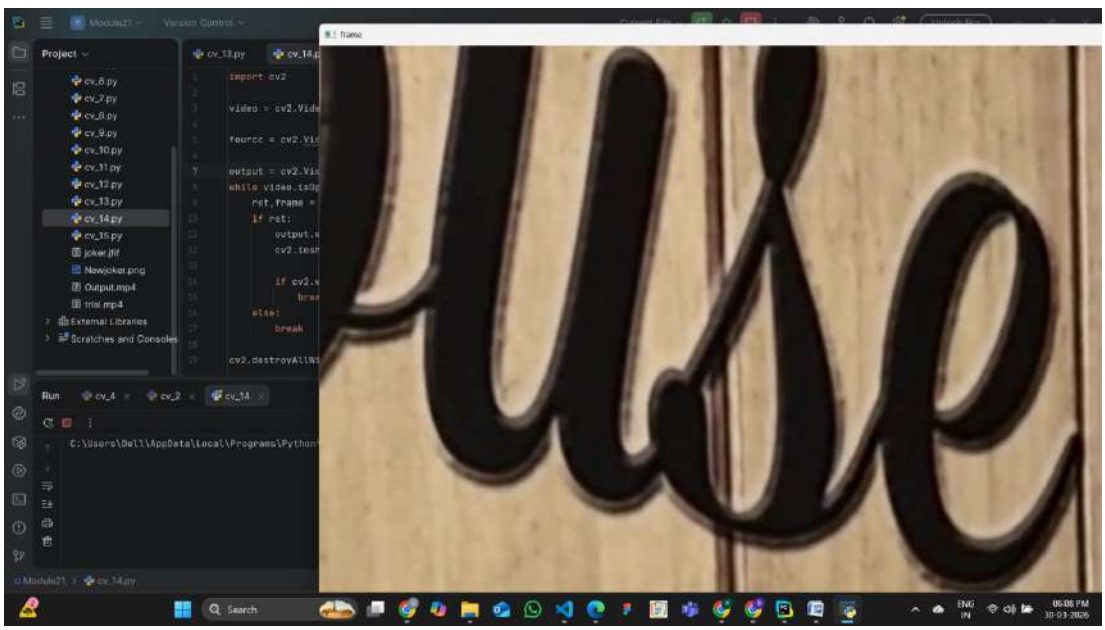
13. cv_13.py

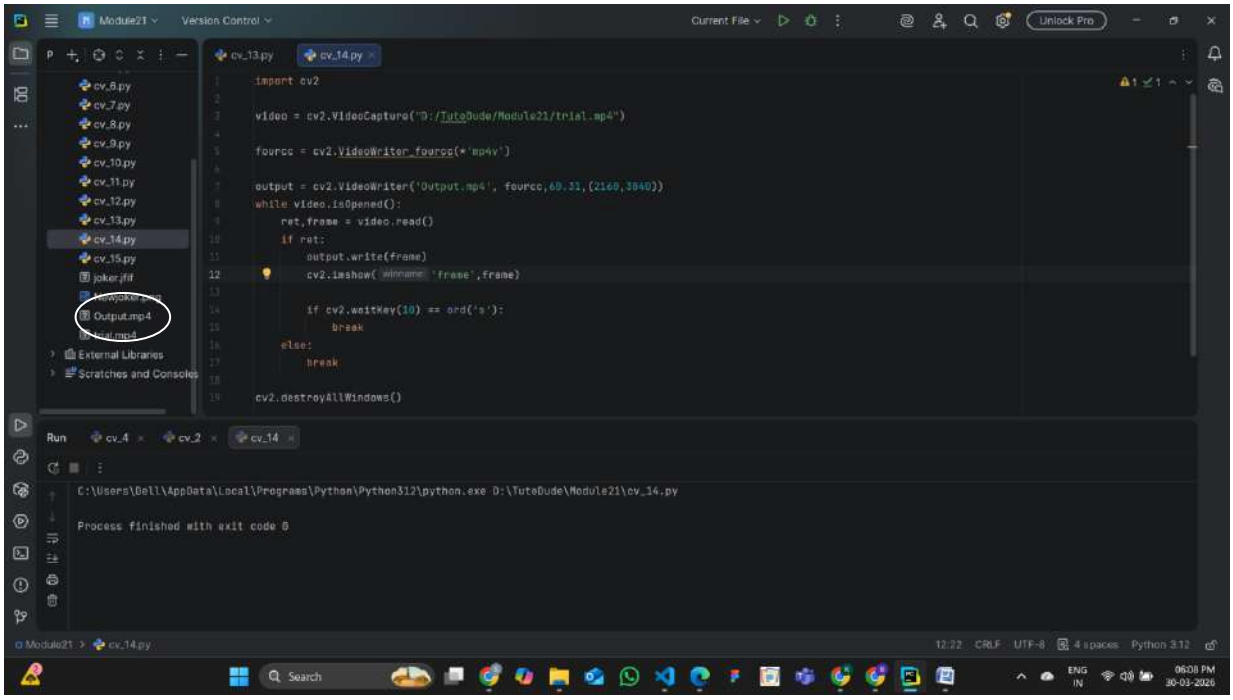
```
import cv2

video = cv2.VideoCapture("D:/TuteDude/Module21/trial.mp4")

while video.isOpened():
    _, frame = video.read()
    frame= cv2.resize(frame,(500,700))
    cv2.imshow("frame",frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cv2.destroyAllWindows()
```

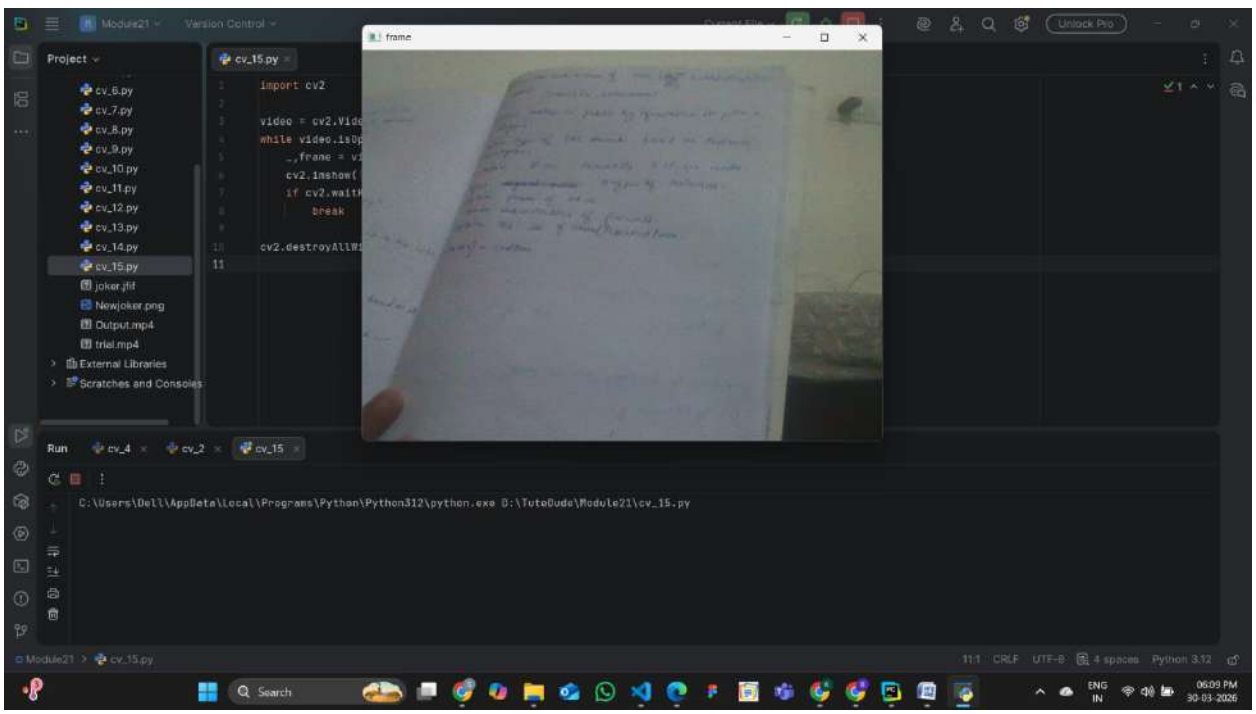


14. cv_14.py

```
import cv2

video = cv2.VideoCapture("D:/TuteDude/Module21/trial.mp4")

fourcc = cv2.VideoWriter fourcc(*'mp4v')

output = cv2.VideoWriter('Output.mp4', fourcc,60.31,(2160,3840))
while video.isOpened():
    ret,frame = video.read()
    if ret:
        output.write(frame)
        cv2.imshow('frame',frame)

        if cv2.waitKey(10) == ord('s'):
            break
    else:
        break

cv2.destroyAllWindows()
```




import cv2
video = cv2.VideoCapture("D:/TuteDude/Module21/trial.mp4")
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
output = cv2.VideoWriter('Output.mp4', fourcc,60.31,(2160,3840))
while video.isOpened():
ret,frame = video.read()
if ret:
output.write(frame)
cv2.imshow('frame',frame)
if cv2.waitKey(10) == ord('s'):
break
else:
break
cv2.destroyAllWindows()
C:\Users\Dell\AppData\Local\Programs\Python\Python312\python.exe D:\TuteDude\Module21\cv_14.py
Process finished with exit code 0

15. cv_15.py

```
import cv2

video = cv2.VideoCapture(0) # 0 means your web cam
while video.isOpened():
    _,frame = video.read()
    cv2.imshow('frame',frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cv2.destroyAllWindows()
```